Algorithm for file updates in Python

Name: Dhyanam
Role: Cyber Security Training
Year: 2026

Project description

Developed a Python script to automate IP allow list management for healthcare networks. The program cross-checks authorized IPs against a remove list, removes unauthorized entries, and updates the file to ensure secure access to patient data. This reduced manual errors and strengthened compliance with data protection standards.

Open the file that contains the allow list

```python
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"
# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]
# First line of `with` statement
with open(import_file,"r") as file:
```

- In the above screenshot we have assigned our file ***allow_list.txt*** which contains the allow list of the IP addresses to ***import_file*** variable.
- Later, we have also defined a list which contains IP addresses that are no longer allowed to access the restricted information.
- The last line uses python's ***open()*** function to access the specified file in read-only mode ***("r")***. There are two additional modes which can be used with the ***open()*** function which are: write mode ***("w")*** which basically replaces the contents or creates a new file and Append mode ***("a")*** which adds content to the end of the file.
- The ***with*** keyword creates a context manager, which ensures that the file is properly opened for use and then automatically closed once the block of code is finished.
- This approach improves code safety and readability by preventing resource leaks and handling file closure without explicitly requiring ***file.close()***.

Read the file contents

```
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"
# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]
# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:
  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
  ip_addresses=file.read()
# Display `ip_addresses`
print(ip_addresses)
```

- Here, we have read the contents of the file using the ***read()*** method which basically converts the file content into a string which is assigned to string variable ***ip_addresses***.

Convert the string into a list

```
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"
# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]
# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:
  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
  ip_addresses = file.read()
# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()
# Display `ip_addresses`
print(ip_addresses)
```

- Here, we have used the ***.split()*** method to convert the string which was our file content into a list format so that we can better perform the comparison operation to find out all the ip addresses which are there in the remove_list.
- The syntax of ***.split()*** method: ***String.split(separator)***
- This splits as a specified separator which can be **‘,’** (comma) or **“ ”** (Whitespace).
- If there is no argument then it splits at whitespaces which can be (spaces, tabs or newlines).

Iterate through the allow list

```python
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# Loop through `ip_addresses`

for element in ip_addresses:
    print(element)
```

- We are using a for loop to iterate through the list ***ip_addresses*** where ***element*** is tracking each element of the list.

Remove IP addresses that are on the remove list

```
# Assign `import_file` to the name of the file
import_file = "allow_list.txt"
# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.
remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]
# Build `with` statement to read in the initial contents of the file
with open(import_file, "r") as file:
  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`
  ip_addresses = file.read()
# Use `.split()` to convert `ip_addresses` from a string to a list
ip_addresses = ip_addresses.split()
# Loop through `ip_addresses`
for element in ip_addresses:
  # If current element is in `remove_list`,
    if element in remove_list:
        # then current element should be removed from `ip_addresses`
        ip_addresses.remove(element)
# Display `ip_addresses`
print(ip_addresses)
```

- We will first iterate through our allow list and compare each element against the elements in the remove list and if they match then we will use the ***remove()*** method of the list to remove that particular element from the list.

Update the file with the revised list of IP addresses

```
# Assign `import_file` to the name of the file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addresses that are no longer allowed to access restricted information.

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.58.57"]

# Build `with` statement to read in the initial contents of the file

with open(import_file, "r") as file:

  # Use `.read()` to read the imported file and store it in a variable named `ip_addresses`

  ip_addresses = file.read()

# Use `.split()` to convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()
`
# Loop through `ip_addresses`

for element in ip_addresses:

  # If current element is in `remove_list`,

    if element in remove_list:

        # then current element should be removed from `ip_addresses`

        ip_addresses.remove(element)

# Convert `ip_addresses` back to a string so that it can be written into the text file

ip_addresses = " ".join(ip_addresses)

# Build `with` statement to rewrite the original file

with open(import_file,"w") as file:

  # Rewrite the file, replacing its contents with `ip_addresses`
    file.write(ip_addresses)
```

- So after removing all the elements that were in the remove list from our allow list now we will use the ***.join()*** method to again convert the list back to string so that we can write our updated information to the file.
- Syntax of ***.join()*** method: ***"Separator".join(list)***
- This concatenates list elements into one string and inserts the chosen separator between elements.
- We opened the file with ***"Write"*** mode and passed our string ***ip_addresses*** as an argument which led to all the data getting written to our file.

Summary

So finally we were able to update the file contents of the allow_list.txt to remove all the ip addresses which no longer have access to the restricted information using python code.